- Procedures must be used, for example to:
  - set up an empty array at the start of each game
  - display the contents of the array
  - enter a player's move – this could use a parameter for `O` or `X`
  - check for a winner
- Your program must be fully commented.
- Your answer to 12 will help you to start your program.

9 Databases

1 a
- table: a collection of records …
- … where each record contains data about a similar item, e.g. a student.
- record: a collection of fields about the same item …
- … there are the same number of fields in each record.
- field: a single piece of data …
- … that is given a data type.

b Four from:
- text/alphanumeric – e.g. a description of a part
- character – e.g. S, M or L for the size of a part
- Boolean – e.g. true or false, sold or not
- integer – e.g. number of parts in stock
- real – e.g. price of an item 12.99
- date/time – e.g. a date such as 18/12/2020

2 a

Field 1: Type
Data type: text
Reason: as the name given to the type of ice cream is a series of characters
Sample: choc ice
Field 2: Flavour
Data type: text
Reason: as the name given to the flavour of ice cream is a series of characters
Sample: vanilla
Field 3: Size
Data type: text
Reason: as the size of the ice cream is stored as a word
Sample: Small
Field 4: NumberInStock
Data type: integer
Reason: as the number in stock will be a whole number
Sample: 56

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Field 5: ReOrderLevel

Data type: integer

Reason: as the re-order level will be a whole number

Sample: 24

b i A primary key is required to uniquely identify each record.

ii All of the fields could be repeated in another record.

iii Name: IceCreamID

Date type: text

Reason: so the field can contain letters and digits

Sample: IC007

c

ICECREAM

Field Name	Data Type	
IDIceCream	Short Text	8 characters only
Type	Short Text	
Flavour	Short Text	
Size	Short Text	
NumberInStock	Number	
ReOrderLevel	Number	

d

ICECREAM

IDIceCream	Type	Flavour	Size	NumberInStock	ReOrderLevel
IC000001	Lolly	Strawberry	Large	35	20
IC000002	Lolly	Raspberry	Small	14	20
IC000003	Choc Ice	Vanilla	Medium	60	30
IC000004	Choc Ice	Chocolate	Medium	30	20
IC000005	Tub	Vanilla	Small	0	10
IC000006	Tub	Vanilla	Large	15	10

3 a Structured Query Language (SQL) is the standard query language …

… for writing scripts to obtain information from a database.

b i
- SELECT: Fetches specified fields (columns) from a table …
- … queries always begin with `SELECT`

ii
- FROM: Identifies the table to use …
- … queries always include `FROM`.

iii
- WHERE: Includes only records (rows) in a query that match a given condition …
- … `WHERE` is an optional command.

iv
- SUM: Adds the values in a specified field …
- … must be an integer or real field.

c `ORDER BY` and `COUNT`

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d **Query 1**

```
SELECT TYPE, Size
FROM ICECREAM
WHERE ((NumberInStock)>0)
ORDER BY NameIceCream;
```

Query 2

```
SELECT SUM(NumberInStock)
FROM ICECREAM;
```

Query 3

```
SELECT Count(NumberInStock)
FROM ICECREAM
WHERE(NumberInStock < ReOrderLevel);
```

4 **a** **i**
- Licence …
- … as it uniquely identifies the teacher.

ii

Name:	text
Title:	text
Licence:	text
Gender:	char or Boolean
Subject:	text
Class:	text

iii Learner's own database

b **i**
- Name …
- … so that any copying errors during data entry can be identified.

ii
- Any of Gender, Title, Licence, Subject, Class could be validated …
- … as there are only a small number of different entries allowed for each.

iii Learner's own validation

c **i** Miss Sing

Mr Ling

ii
```
SELECT Title, Name
  FROM TEACHER
  WHERE Subject = "Science"
  ORDER BY Name;
```

iii
```
SELECT Title, Name
  FROM TEACHER
  WHERE (Subject = "Science" OR Subject = "Mathematics")
  ORDER BY Name;
```

iv
```
SELECT COUNT (Subject)
  FROM TEACHER
  WHERE (Subject = "Mathematics");
```

v Learner's own check

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